BRA - Beta Ray Spectroscopy

Signature Sheet

Student's Name _____ Partner's Name _____

Suggested reading to start with (see the end of the manual for more):

1. 111B BRS Manual
2. ...
3. 111B **Beta-Ray Spectroscopy Video**. Note: In order to view the private Youtube videos hosted by the university, you must be signed into your berkeley.edu Google account.

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. What are beta rays? How are they produced?
2. How is radioactive decay related to nuclear binding energy?
3. What is the half-life of Cs-137 and what are the decay modes of this nucleus? Sketch the two beta-decay distributions of Cs-137. Qualitatively explain their general shapes and relative sizes. Indicate the maximum energy of each distribution.
4. What is the "internal conversion process" and how does atomic electronic binding play a role in the electron's final kinetic energy?
5. What is a Fermi-Kurie plot, and why do you need to use it?
6. What does the beta-ray spectrometer measure, and how does it measure it?
7. Identify and describe the operation of each of the pieces of experimental apparatus: source, coils, preamp, amp, SCA, etc.

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. On day 2 of this lab, you should have taken several quick spectra and determined an appropriate setting of the discriminator's baseline parameter. Demonstrate and show the spectra to an instructor and ask for a signature.

Staff Signature _____ Date _____

Completed by day 2 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. LabView

   Staff Signature _____

2. Scope Image

   Staff Signature _____

3. Statistical Fluctuations

   Staff Signature _____

4. Hysteresis

   Staff Signature _____

5. Combined Fermi-Kurie Plot

   Staff Signature _____